

Dependence Uncertainty

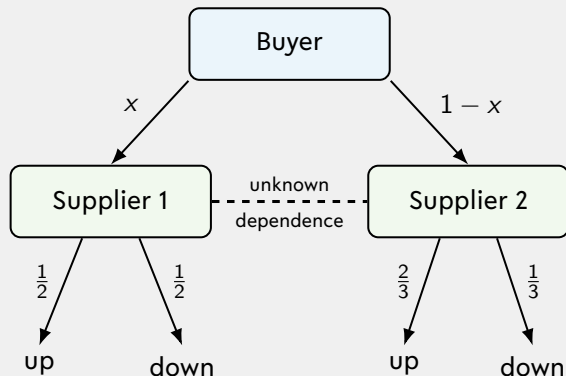
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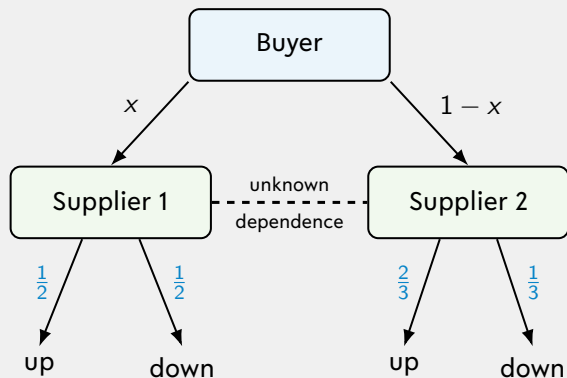
IFORS Vienna, July 2026

Dual-sourcing procurement with disruption



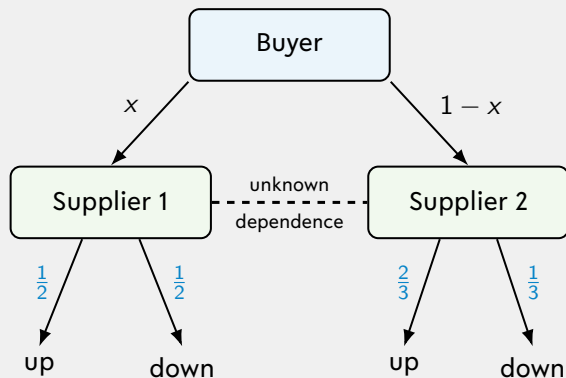
- Sourcing share x to Supplier 1.
- Known individual disruption rates (audits, service histories).
- Dual sourcing enhances reliability and resilience, in case of common shocks (natural disaster, geopolitics).
- Optimal sourcing depends on the unknown joint disruption structure.

Dual-sourcing procurement with disruption



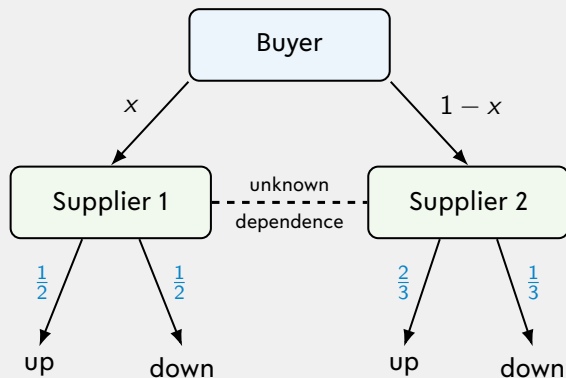
p_{ind}	u_2	d_2
u_1	$\frac{1}{3}$	$\frac{1}{6}$
d_1	$\frac{1}{3}$	$\frac{1}{6}$

Dual-sourcing procurement with disruption



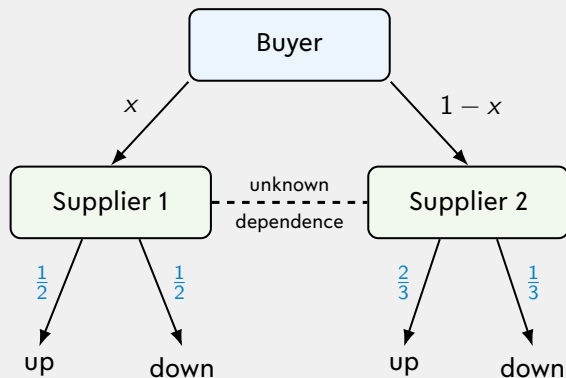
p	u_2	d_2
u_1	$\frac{1}{3} + a$	$\frac{1}{6}$
d_1	$\frac{1}{3}$	$\frac{1}{6}$

Dual-sourcing procurement with disruption



p	u_2	d_2
u_1	$\frac{1}{3} + a$	$\frac{1}{6} - a$
d_1	$\frac{1}{3} - a$	$\frac{1}{6} + a$

Dual-sourcing procurement with disruption



p	u_2	d_2
u_1	$\frac{1}{3} + a$	$\frac{1}{6} - a$
d_1	$\frac{1}{3} - a$	$\frac{1}{6} + a$

- $a \in [-\frac{1}{6}, \frac{1}{6}]$
- p_{ind} for $a = 0$
- $p = p_{\text{ind}} + q$, $q = \begin{pmatrix} a & -a \\ -a & a \end{pmatrix}$

Consider a finite state space $\Omega = \Omega_1 \times \dots \times \Omega_n$ with prescribed $p_i \in \Delta(\Omega_i)$.

Definition

The *dependence uncertainty set* is

$$\mathcal{P} = (\Omega; p_1, \dots, p_n) := \{p \in \Delta(\Omega) \mid p|_{\Omega_i} = p_i \forall i \in \{1, \dots, n\}\}.$$

$\mathcal{P}$ is a compact convex polytope with $p_{\text{ind}} := p_1 \otimes \dots \otimes p_n \in \mathcal{P}$.

► extreme points

► dimension

Definition

The set of *dependence structures* is

► copulae

$$Q = Q(\Omega) := \{q: \Omega \rightarrow \mathbb{R} \mid \sum_{\omega_{-i}} q(\omega, \omega_{-i}) = 0 \forall \omega_i \in \Omega_i\}.$$

Definition

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Proposition

Every $p \in \mathcal{P}$ can uniquely be written as $p = p_{\text{ind}} + q$ with $q \in Q$. Furthermore,

$$\mathcal{P} = \{p_{\text{ind}} + q \mid q \in Q\} \cap \Delta(\Omega).$$

Consider the set $\mathcal{F}$ of acts $f: \Omega \rightarrow X$ (Anscombe and Aumann, 1963).
Today, $X = \mathbb{R}$ and risk-neutrality, i.e., $u(x) = x$.

Preferences and Dependence Uncertainty

Consider the set $\mathcal{F}$ of acts $f: \Omega \rightarrow X$ (Anscombe and Aumann, 1963).
Today, $X = \mathbb{R}$ and risk-neutrality, i.e., $u(x) = x$.

We propose to say a preference relation $\succeq$ on $\mathcal{F}$ reflects *dependence uncertainty*, if there are marginals $p_i \in \Delta(\Omega_i)$ and an act-dependent *ambiguity index* $\alpha: \mathcal{F} \rightarrow [0, 1]$ such that $\succeq$ is represented by the functional

$$V(f) = \alpha(f) \min_{p \in \mathcal{P}} \int_{\Omega} f \, dp + (1 - \alpha(f)) \max_{p \in \mathcal{P}} \int_{\Omega} f \, dp,$$

where $\mathcal{P} = \mathcal{P}(\Omega; p_1, \dots, p_n)$.

Definition

An act $f \in \mathcal{F}$ is called *dependence neutral*, if for all $p, p' \in \Delta(\Omega)$ with $p|_{\Omega_i} = p'|_{\Omega_i}$ for all $i \in \{1, \dots, n\}$ it holds that $\int_{\Omega} f \, dp = \int_{\Omega} f \, dp'$. Their set is $\mathcal{F}^{\text{dn}}$.

Equivalently, $\int_{\Omega} f \, dq = 0$ for all $q \in Q$.

f_1	u_2	d_2
u_1	x	x
d_1	0	0

$$f_1 \in \mathcal{F}_1$$

f_2	u_2	d_2
u_1	$1 - x$	0
d_1	$1 - x$	0

$$f_2 \in \mathcal{F}_2$$

$f_1 + f_2$	u_2	d_2
u_1	1	x
d_1	$1 - x$	0

$$f_1 + f_2 \in \mathcal{F}^{\text{dn}}$$

- A1. $\succsim$ is complete, transitive and non-trivial.
- A2. ($\mathcal{F}^{\text{dn}}$ -Archimedean) For all $g \in \mathcal{F}$ and all $f, h \in \mathcal{F}^{\text{dn}}$, if $f \succ g \succ h$, then there exist $\lambda, \kappa \in (0, 1)$ such that $\lambda f + (1 - \lambda)h \succ g \succ \kappa f + (1 - \kappa)h$.

- A3. ($\mathcal{F}^{\text{dn}}$ -Independence) For all $f, g, h \in \mathcal{F}^{\text{dn}}$, $\lambda \in (0, 1)$

$$f \succsim g \iff \lambda f + (1 - \lambda)h \succsim \lambda g + (1 - \lambda)h.$$

- A4. ($\mathcal{F}^{\text{dn}}$ -Dominance) For all $f \in \mathcal{F}$, $g \in \mathcal{F}^{\text{dn}}$

$$f(\omega) \succeq (\preceq) g(\omega) \forall \omega \implies f \succsim (\preceq) g.$$

Theorem

A preference relation $\succsim$ on $\mathcal{F}$ fulfills satisfies **Axioms 1–4** if and only if there exist an ambiguity index $\alpha: \mathcal{F} \rightarrow [0, 1]$ and $p_1, \dots, p_n \in \Delta(\Omega_1), \dots, \Delta(\Omega_n)$ such that $\succsim$ is represented by the functional

$$V(f) = \alpha(f) \min_{p \in \mathcal{P}} \int_{\Omega} f \, dp + (1 - \alpha(f)) \max_{p \in \mathcal{P}} \int_{\Omega} f \, dp,$$

where $\mathcal{P} := \mathcal{P}(\Omega; p_1, \dots, p_n)$.

Compatibility to other uncertainty representation

- Maxmin expected utility (MEU) (Gilboa and Schmeidler, 1989, JME)

$$V_{\text{MEU}}(f) = \min_{p \in \mathcal{C}} \int_{\Omega} f \, dp \quad \text{where } \mathcal{C} \subseteq \mathcal{P}.$$

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- Variational preferences (V) (Maccheroni, Marinacci, and Rustichini, 2006, ECMA; Hansen and Sargent, 2001, AER)

$$V_{\text{VP}}(f) = \min_{p \in \Delta\Omega} \left\{ \int_{\Omega} f \, dp - c(p) \right\} \quad \text{where } c(p) = \infty \text{ on } \Delta(\Omega) \setminus \mathcal{P}.$$

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- Smooth ambiguity (Klibanoff, Marinacci, and Mukerji, 2005, ECMA)

$$V_{\text{smooth}}(f) = \int_{\Delta(\Omega)} \phi \left(\int_{\Omega} f \, dp \right) \, d\mu \quad \text{where } \text{supp}(\mu) \subseteq \mathcal{P}.$$

Dependence sensitive act

f	u_2	d_2
u_1	1	
d_1		

Dependence sensitive act

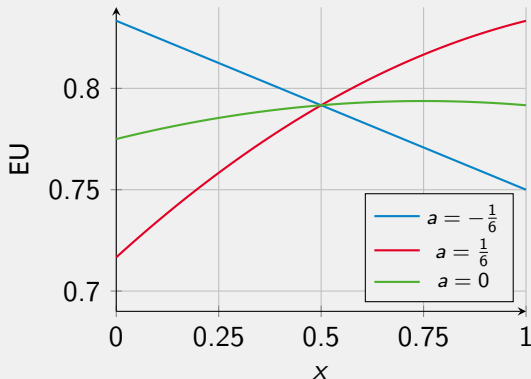
f	u_2	d_2
u_1	1	$x + \frac{1}{2}(1 - x)$
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Dependence sensitive act

f	u_2	d_2
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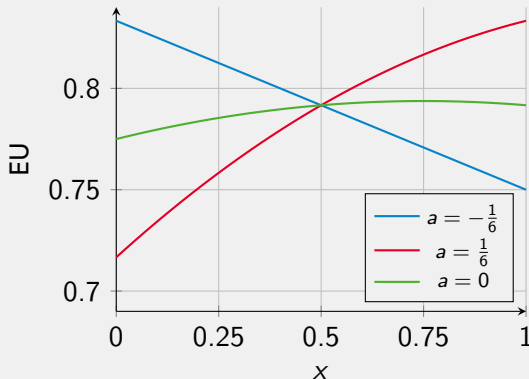


► state-wise utility

Dependence sensitive act

f	u_2	d_2
u_1	1	$x + \frac{1}{2}(1 - x)$
d_1	$\frac{1}{2}x + (1 - x)$	$\frac{3}{20} + \frac{4}{5}x - \frac{1}{5}x^2$

- $a = -\frac{1}{6} \rightarrow x^* = 0$
- $a \in \{0, \frac{1}{6}\} \rightarrow x^* = 1$
- $a \in [-\frac{1}{6}, \frac{1}{6}] \& \text{ MEU} \rightarrow x^* = \frac{1}{2}$



► state-wise utility

Humans neglect dependence

- when making financial decisions (Eyster and Weizsäcker, 2010; Eyster and Weizsäcker, 2016; Kallir and Sonsino, 2009),
- when forming beliefs (Enke and Zimmermann, 2019, REStud).

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Formal representation

$$V(f) = \int_{\Omega} f \, dp_{\text{ind}}.$$

- We make dependence neglect testable by an axiom!

A5. (Subspace Separation) For all $f_i, g_i \in \mathcal{F}_i$, $y \in X$, non-null $E_{-i} \subseteq \Omega_{-i}$,

$$f_i \succcurlyeq g_i \iff f_{i[E_{-i}]}y \succcurlyeq g_{i[E_{-i}]}y.$$

<table style="border-collapse: collapse; margin: 0 auto;"> <tr> <td style="border-right: 1px solid black; padding: 5px;">f_1</td> <td style="border-right: 1px solid black; padding: 5px;">u_2</td> <td style="padding: 5px;">d_2</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 5px;">u_1</td> <td style="border-right: 1px solid black; padding: 5px;">1</td> <td style="padding: 5px;">1</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 5px;">d_1</td> <td style="border-right: 1px solid black; padding: 5px;">0</td> <td style="padding: 5px;">0</td> </tr> </table>	f_1	u_2	d_2	u_1	1	1	d_1	0	0	$\succcurlyeq$ $\iff$	<table style="border-collapse: collapse; margin: 0 auto;"> <tr> <td style="border-right: 1px solid black; padding: 5px;">g_1</td> <td style="border-right: 1px solid black; padding: 5px;">u_2</td> <td style="padding: 5px;">d_2</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 5px;">u_1</td> <td style="border-right: 1px solid black; padding: 5px;">0.8</td> <td style="padding: 5px;">0.8</td> </tr> <tr> <td style="border-right: 1px solid black; padding: 5px;">d_1</td> <td style="border-right: 1px solid black; padding: 5px;">0.4</td> <td style="padding: 5px;">0.4</td> </tr> </table>	g_1	u_2	d_2	u_1	0.8	0.8	d_1	0.4	0.4
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Theorem

If $\succsim$ is MEU with prior set $\mathcal{C} \subseteq \mathcal{P}(\Omega; p_1, \dots, p_n)$, Axiom 5 holds if and only if $\succsim$ is SEU with respect to the independent product $p_{\text{ind}} = p_1 \otimes \dots \otimes p_n$.

Definition

We say that $\succsim$ is *more dependence averse* than $\succsim'$ if for all $f \in \mathcal{F}$ and $g \in \mathcal{F}^{\text{dn}}$

$$f \succsim g \implies f \succsim' g.$$

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Definition

The *dependence premium* of $f \in \mathcal{F}$ is the value $\text{DP}(f)$ satisfying

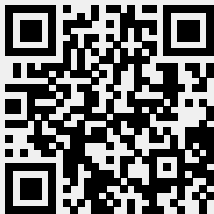
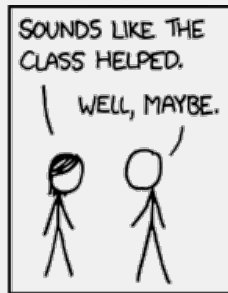
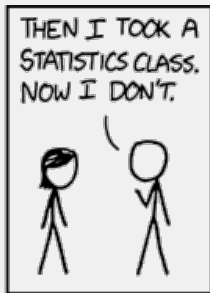
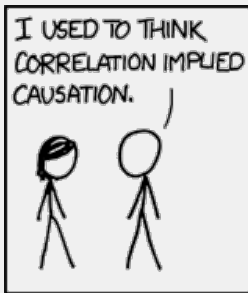
$$\int_{\Omega} f - \text{DP}(f) \, dp_{\text{ind}} = V(f)$$

Proposition

For dependence uncertainty preferences, $\succsim$ is more dependence averse than $\succsim'$ iff one of the following statements hold:

- (i) $p_i = p'_i$ for all $i \in \{1, \dots, n\}$ and $\alpha(f) \geq \alpha'(f)$ for all $f \in \mathcal{F} \setminus \mathcal{F}^{\text{dn}}$.
- (ii) $\text{DP}(f) \geq \text{DP}'(f)$ for all $f \in \mathcal{F}$.
- (iii) $\mathcal{C}' \subseteq \mathcal{C}$ (MEU) (iv) $c^* \leq c'^*$ (VP) (v) $\phi = h \circ \phi'$, h concave (smooth)

► plots



Bauch and Hartmann, 2026

Thank You!

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-  Bauch, Gerrit and Lorenz Hartmann (2026). *Dependence uncertainty: a decision-theoretic approach*. arXiv: 2503.13416 [econ.TH].
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-  Eyster, Erik and Georg Weizsäcker (2010). *Correlation neglect in financial decision-making*. DIW Berlin Discussion Paper.
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-  Hansen, Lars Peter and Thomas J Sargent (2001). “Robust control and model uncertainty”. In: *American Economic Review* 91.2, pp. 60–66.
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-  Maccheroni, Fabio, Massimo Marinacci, and Aldo Rustichini (2006). “Ambiguity aversion, robustness, and the variational representation of preferences”. In: *Econometrica* 74.6, pp. 1447–1498. ISSN: 00129682, 14680262. URL: <http://www.jstor.org/stable/4123081>.
-  Nelsen, Roger B (2006). *An introduction to copulas*. Springer. ISBN: 978-0-387-28659-4. DOI: 10.1007/0-387-28678-0.

Theorem

The following are equivalent for a probability $p \in \mathcal{P}$ and any divergence $D(\cdot \parallel \cdot)$ with infinitely decreasing slope in 0.

- (i) p is an extreme point of $\mathcal{P}$.*
- (ii) p is maximally zero.*
- (iii) p is a strict local maximizer of the divergence $D(\cdot \parallel p_{\text{ind}})$.*

E.g., φ -divergence: $\varphi'(0) = -\infty$. Examples: α -divergence for $\alpha \in [0, 1]$, including (generalized, reversed) Kullback-Leibler divergence, Jeffrey's divergence, Jensen-Shannon divergence, squared Hellinger distance, Neyman χ^2 -divergence (aka reverse Pearson).

Rectangular operations are dependence structures q with support on a rectangle:

$$\begin{array}{ccc}
 (\omega_i^*, \omega_{-i}^*) & \xleftarrow{a} & (\omega_i^*, \omega_{-i}^\dagger) \\
 & \text{X} & \\
 (\omega_i^\dagger, \omega_{-i}^*) & \xrightarrow{a} & (\omega_i^\dagger, \omega_{-i}^\dagger)
 \end{array}$$

Proposition

Rectangular operations form a spanning system of Q . Setting $m_i := \# \text{supp}(p_i)$, the dimension of $\mathcal{P}$ is equal to $\prod_{i=1}^n m_i - \sum_{i=1}^n (m_i - 1) - 1$.

- Correlation is often measured by a *copula* in statistics, cf. Nelsen, 2006.
- A copula is a joint distribution of $[0, 1]^n$ with *uniform* marginals.
- Copula theory works only for real-valued random variables
 - Dependence structures work on arbitrary state spaces.
 - Copulae can indeed be identified as special dependence structures.

▸ dependence structures

Dependence Structures and Copulae — II

C	$\frac{2}{3}$	1
$\frac{1}{4}$	$\frac{1}{6} + x$	$\frac{1}{4}$
1	$\frac{2}{3}$	1

$C - C_{\text{ind}}$	$\frac{2}{3}$	1
$\frac{1}{4}$	x	0
1	0	0

$p_{\text{ind}} + q$	$\frac{2}{3}$	1
$\frac{1}{4}$	$\frac{1}{6} + x$	$\frac{1}{12} - x$
1	$\frac{1}{2} - x$	$\frac{1}{4} + x$

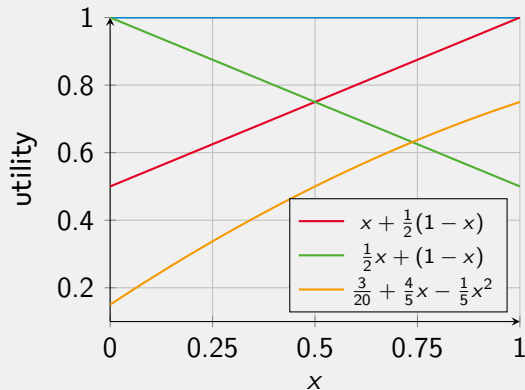
q	$\frac{2}{3}$	1
$\frac{1}{4}$	x	$-x$
1	$-x$	x

A copula C and its corresponding dependence structure q for $\Omega_1 = \{\frac{1}{4}, 1\}$, $\Omega_2 = \{\frac{2}{3}, 1\}$ with $x \in [-\frac{1}{6}, \frac{1}{12}]$.

► dependence structures

Dependence sensitive act – State-wise Utility

f	u_2	d_2
u_1	1	$x + \frac{1}{2}(1 - x)$
d_1	$\frac{1}{2}x + (1 - x)$	$\frac{3}{20} + \frac{4}{5}x - \frac{1}{5}x^2$



► back

Dependence Premia for Different Preferences

